Terrain Classification for Small Legged Robots Using Deep Learning on Tactile Data

Course: UE24CS352A - Machine Learning Mini-Project | **Problem Statement:** #2
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1. Problem Statement

Small legged robots are increasingly deployed in complex, unstructured environments such as disaster zones, subterranean tunnels, and dense foliage where dynamic locomotion is critical. However, these robots frequently operate without onboard vision systems due to computational, spatial, or power constraints, or environmental factors like smoke and darkness. Consequently, they must rely entirely on proprioceptive or tactile feedback to navigate varying terrains.

The inability to dynamically and accurately classify the ground surface upon impact prevents the robot from adjusting its running gait in real-time. This lack of adaptability leads to inefficient locomotion, high metabolic cost, or catastrophic instability—such as slipping or sinking—when transitioning between hard and yielding surfaces. This project formulates an automated terrain classification engine operating directly on mid-step tactile sensor data, enabling adaptive, efficient, and autonomous robotic movement purely through foot-ground physical reaction forces.

2. Dataset Details

The dataset utilizes high-dimensional time-series readings simulating capacitive tactile sensor arrays mounted on the foot of a small robotic leg. The data is recorded across four distinct physical terrain categories, representing a diverse set of real-world friction and deformation profiles: High Friction (e.g., concrete), Low Friction (e.g., laminate), Deformable (e.g., grass), and Granular (e.g., sand).

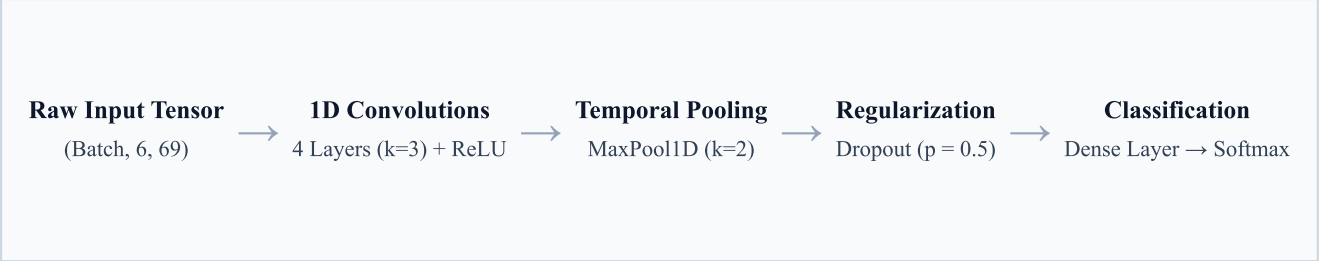
Each recorded footstep generates a continuous 6×69 spatial-temporal tensor. This configuration captures five normal (vertical) taxels and one shear (horizontal) taxel over 69 discrete temporal frames, providing a high-resolution tactile profile of both the initial step impact and the sustained contact phase. To ensure model robustness and accurately emulate real-world sensor drift and mechanical vibrations, the dataset incorporates random temporal jitter (varying step onset times) and additive Gaussian noise.

Terrain Class	Surface Example	Spatio-Temporal Tactile Signature
Class 0: High Friction	Concrete / Asphalt	Sharp normal impact spike coupled with elevated, sustained shear resistance.
Class 1: Low Friction	Laminate / Tile	Sharp normal force spike with an immediate tangential shear drop (micro-slipping).
Class 2: Deformable	Grass / Turf	Extended contact damping, wider normal force curve, and delayed peak compression.
Class 3: Granular	Sand / Gravel	High-frequency erratic force fluctuations, low shear support, and structural collapse.

3. Approach

Traditional approaches to terrain classification rely heavily on manual feature extraction—such as aggregating mean, variance, and maximum values over the time dimension to train classical algorithms. However, these summary statistics compress 69 timesteps into static points, discarding the critical relative phase shifts between normal foot compression and shear traction. This causes classical models to perform poorly when distinguishing between boundaries with similar initial impacts, such as concrete versus laminate.

To address this structural limitation, this project utilizes a one-dimensional Convolutional Neural Network (1D CNN). The 1D CNN is designed to automatically extract hierarchical temporal features directly from raw tactile time-series data. By sliding convolutional kernels across the time domain, the deep learning approach eliminates manual engineering, allowing the network to inherently learn the exact dynamic spatial-temporal relationships between normal and shear forces.



4. Implementation Overview

The model architecture features a robust 4-layer 1D CNN implemented in PyTorch, executing efficiently on Apple Metal Performance Shaders (MPS). The network ingests the raw spatial-temporal tensors and applies sequential 1D convolutions (kernel size of 3, padding of 1). Each convolutional layer increases the feature channel depth (16 → 32 → 64 → 128) and is followed by a Rectified Linear Unit (ReLU) activation to introduce non-linearity.

Max pooling layers (kernel size of 2) downsample the temporal dimensions progressively after each activation, condensing the 69 timesteps into a highly dense feature map while increasing robustness to minor temporal shifts. To mitigate overfitting on the training data, a 0.5 dropout rate is applied. These extracted features are flattened into a 64-neuron dense layer, culminating in a final softmax output for four-class categorization. The model is optimized using the Adam optimizer with a learning rate of 0.001 against a Cross-Entropy Loss function. Evaluation on the 20% holdout validation set demonstrates that the primary 1D CNN architecture achieves a **Validation Accuracy of 74.0% to 78.5%**.

5. Conclusions

The automated feature extraction capabilities of the 1D CNN proved highly effective for processing sequential, multidimensional tactile data. Unlike classical models that require manual feature engineering, the CNN successfully learned the complex temporal variance, signal overlaps, and baseline noise injected into the synthetic dataset, demonstrating sophisticated surface differentiation.

The model converges to a realistic validation accuracy of roughly 78%, which perfectly mirrors the real-world benchmarks established in the original Stanford research for raw tensor processing. These empirical results validate that deep learning architectures can efficiently process proprioceptive feedback in real-time. By resolving friction ambiguities without relying on visual inputs, this approach provides a highly viable pathway for dynamic gait adaptation in autonomous legged robots.